شهادة إتمام تدريب

CERTIFICATE OF COMPLETION

تبارك الأكاديمية السعودية الرقمية لـ

Mathayil Fahad Alotaibi

مثايل فهد العتيبي

On successfully completing **Generative AI LLM Specialization Bootcamp** in a journey that lasted **13 weeks** From **24 August 2025** to **20 November 2025**

Issue date **20 November 2025**

إتمام معسكر **همة للذكاء الاصطناعي التوليدي نماذج اللغات الكبيرة** بنجاح بعد رحلة استمرت لمدة **13 أسبوع** في الفترة من **1 ربيع الأول 1447** الى **29 جمادى الأول 1447**

تم إصدارها بتاريخ **29 جمادى الأول 1447**

WISHING YOU A CONTINOUS SUCCESS

مع تمنياتنا بدوام التوفيق والنجاح

Amal A. Aljuhaiman

الرئيس التنفيذي للأكاديمية السعودية الرقمية

امل عبدالكريم الجهيمان

Bootcamp Syllabus

WEEK 1

Pre-bootcamp

- Welcome session (meet and greet)
- Bootcamp overview and introduction
- Week 0 self-paced learning preparation guide
- Learning support

WEEK 2

Python for Data Science

- Python syntax, Jupyter notebooks, pandas, NumPy, data types, loops, functions

WEEK 3

Data Wrangling & EDA

- Data wrangling, missing values, visualization (matplotlib/seaborn), basic statistics, correlation

WEEK 4

Introduction to Machine Learning

- Supervised learning, classification/regression, scikit-learn workflow, train-test split, metrics
- Feature scaling, encoding, cross-validation, metrics (precision, recall, F1), hyperparameter tuning

WEEK 5

Deep Learning Fundamentals

- Neural networks, activation functions, backpropagation,
- PyTorch intro, building a simple NLP classifier

WEEK 6

NLP Foundations

- Text preprocessing, tokenization, embeddings(TF-IDF, Word2Vec), sentiment analysis, classification

WEEK 7

Introduction to Transformers & LLMs

- Transformer architecture, Hugging Face models, text generation, summarization, embeddings, prompt engineering

WEEK 8

LLM Fine-tuning

- LoRA(Low-Rank Adaptation), PEFT (Parameter-Efficient Fine-Tuning), instruction tuning, reinforcement learning with human feedback (RLHF), prompt tuning, dataset formatting

Bootcamp Syllabus

WEEK 9

RAG& AI Agent Development

- RAG architecture, vector stores (FAISS, Chroma, Milvus), LangChain/Haystack agents, tool usage, chain-of-thought reasoning

WEEK 10

Capstone Kickoff & Project Design

- Project scoping, team formation, client briefing (if applicable), dataset exploration, planning workflow

WEEK 11

Capstone Development

- Model development, iteration, fine-tuning, integrating retrieval or agents as needed

WEEK 12

Capstone Completion & Demo Day

- Project Wrap-up and Final Client Project Presentation

WEEK 13

Post-bootcamp Certification Preparation

- Certification preparation scheduled afterward to maintain project flow